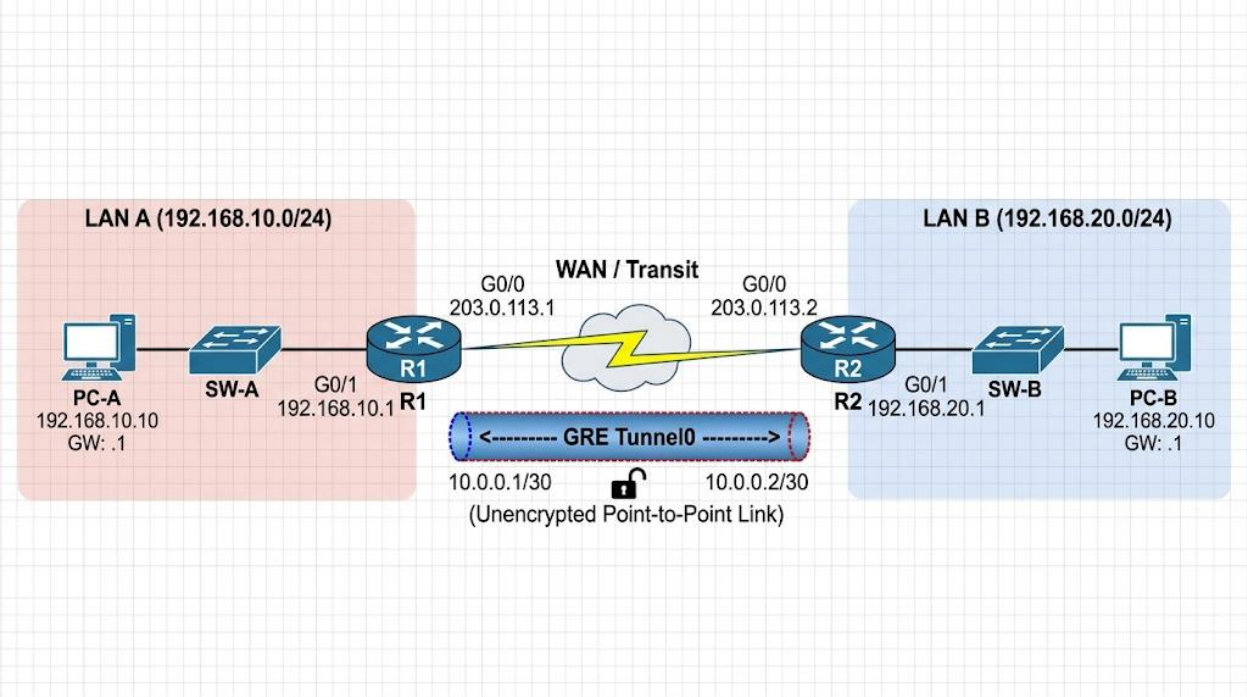
GRE tunnel (no encryption)



Interface and Addressing Table

Node / Device	Interface	IP Address	Subnet Mask	Purpose
R1	GigabitEthernet0/0	203.0.113.1	255.255.255.252	WAN / Underlay Source
R1	GigabitEthernet0/1	192.168.10.1	255.255.255.0	LAN A default gateway
R1	Tunnel0	10.0.0.1	255.255.255.252	GRE Tunnel Interface

Node / Device	Interface	IP Address	Subnet Mask	Purpose
R2	GigabitEthernet0/0	203.0.113.2	255.255.255.252	WAN / Underlay Source
R2	GigabitEthernet0/1	192.168.20.1	255.255.255.0	LAN B default gateway
R2	Tunnel0	10.0.0.2	255.255.255.252	GRE Tunnel Interface
PC-A	NIC	192.168.10.10	255.255.255.0	Host behind R1 (GW: 192.168.10.1)
PC-B	NIC	192.168.20.10	255.255.255.0	Host behind R2 (GW: 192.168.20.1)

Router 1 (R1) Configuration

```
configure terminal

! 1. Physical LAN Interface
interface GigabitEthernet0/1
 description LAN-A
```

```
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit

! 2. Physical WAN Interface
interface GigabitEthernet0/0
 description WAN-to-R2
 ip address 203.0.113.1 255.255.255.252
 no shutdown
 exit

! 3. GRE Tunnel Interface
interface Tunnel0
 description GRE-to-R2
 ip address 10.0.0.1 255.255.255.252
 tunnel source GigabitEthernet0/0
 tunnel destination 203.0.113.2
 tunnel mode gre ip
 no shutdown
 exit

! 4. Route LAN B traffic across the tunnel
ip route 192.168.20.0 255.255.255.0 10.0.0.2
end
write memory
```

Router 2 (R2) Configuration

```
configure terminal

! 1. Physical LAN Interface
interface GigabitEthernet0/1
 description LAN-B
 ip address 192.168.20.1 255.255.255.0
 no shutdown
 exit

! 2. Physical WAN Interface
interface GigabitEthernet0/0
 description WAN-to-R1
 ip address 203.0.113.2 255.255.255.252
 no shutdown
 exit

! 3. GRE Tunnel Interface
interface Tunnel0
 description GRE-to-R1
 ip address 10.0.0.2 255.255.255.252
 tunnel source GigabitEthernet0/0
 tunnel destination 203.0.113.1
 tunnel mode gre ip
 no shutdown
 exit
```

! 4. Route LAN A traffic across the tunnel

ip route 192.168.10.0 255.255.255.0 10.0.0.1

end

Verification Steps

1. **show ip interface brief (Verify Tunnel0 is up/up)**

2. **show interfaces tunnel0 (Check tunnel details and counters)**

3. **show ip route (Ensure the static route to the remote LAN via 10.0.0.x is in the routing table)**

4. **ping 10.0.0.2 (Test point-to-point tunnel reachability from R1)**

5. **ping 192.168.20.10 (Test end-to-end LAN reachability from PC-A or R1 using a LAN source)**